Group Project

This sheet is a guide to the PHY2006 group project in the area of linear algebra and Fourier Transforms.

Contents

1 Aim

The aim of this group project is to gain familiarity with the Fourier transform by filtering sound files in frequency space. A specific goal is to investigate filtering the human voice.

2 Introduction And Background

When sound is recorded the oscillations in air pressure are observed with a microphone and the signal is recorded. This signal can be Fourier transformed into 'frequency space'. The frequency signal is much sparser than the signal from the microphone and some frequencies will have close to zero contribution to the signal.

For telecommunications companies it is highly desirable to compress voice data by removing unecessary frequencies. Good compression does not destroy the quality of the sound, but reduces the data requirement for voice calls enabling more calls to be transmitted than if the time dependent uncompressed signal was transmitted.

3 Method

You will use Python for this project and write the code to process sound files yourselves. Python has libraries with useful functions to load and save sound files and to perform the Fourier transform and inverse Fourier transform. You will be given example code to help you use these libraries. **It is very important that you get this code working as soon as possible in the project.**

The steps that you will need to get working in your code are

- To load data from a `.wav` file.
- To perform the Fourier transform on the sound data.
- To reduce the data in some frequencies to zero.
- To perform the inverse Fourier transform on the modified frequency data.
- To save the processed sound data as a `.wav` file.

Please work on the code early so that any issues/ problems can be ironed out.

4 Goals and assessment

Each group has the following goals

1. Separate the owl and 'beep' sounds in the `owlbeep.wav` file and find and if possible isolate any other sounds in the file.
2. Determine the normal frequency range of the human voice
3. Determine what frequencies can be removed without seriously distorting the human voice

The assessment: each group will

1. make a presentation of maximum 10 minutes to show their findings - it is expected that all group members will contribute to making the presentation more or less equally - please let me know of any issues with this as soon as possible.
2. write a short joint maximum 1 page A4 abstract/summary of their findings, which should be submitted as a **PDF** file and also printed and brought to the presentation.
3. submit their presentation as a **PDF** file

Note that if you would like the first slide of the presentation can be the joint summary; then you would only have to submit the presentation.

5 Additional resources

- The owlbeep.wav sound file will be provided.
- some suitably formatted `.wav` files of the human voice are available at `http://www.voiptroubleshooter.com/open_speech/american.html`

6 FAQ: Frequency Asked Questions

- *How many sound recordings should we attempt to edit and put in our presentation?* You have limited time in your presentation and so have to focus on your key results. Remember you have a maximum of 10 minutes. You have more time in the preparation to listen to and filter different sound files, but you have to manage your time carefully and be careful that you do not spend excessive time working on a large number sound files.

- *How many different voices should we consider?* It is a good idea to consider the voices of different individuals, but you have limited time to work on this and very limited time in the presentation so you have to decide how to do this best within the time available. It is fine to use a few files from the website `http://www.voiptroubleshooter.com/open_speech/american.html`

- *What is required from the summary/presentation; for example, do we just discuss the meaning / importance of our results?* There are three goals listed in the project and you could treat the presentation/1 page summary as opportunities to explain what progress you made heading towards these three goals.

- *We can't get our Python program to work!* Do come to the classes in the computer room and if necessary review videos from these classes.

- *How much do we need to write for the joint summary?* I suggest a short paragraph of 2 or 3 sentences to address each of the three goals listed above. It should be significantly less than 1 page of A4.